

Responses generated by the LLM after the analysis of call paths and its corresponding system tree from cube profiles.

“Path” corresponds to the path of the files as uploaded on sciebo

“Time taken” corresponds to the full time taken from hitting the analyze button till finishing analysis

1. Path: 3/Funwave/scorep_mytdv_scorep_10_trace → profile.cubex

Time taken: ~1 min

1 root; 104 call paths, 10 system tree elements

Analyzing call tree...

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REGION 1

call_path: "Invisible call tree root/funwave_tvd"

severity: 0%

slowest_ranks: 0,1,2,3,4,5,6,7,8,9

reason: All ranks report zero runtime, indicating missing or uniform measurements.

investigate: compare per-rank iteration counts and workload sizes for the listed ranks

fix: redistribute mesh partitions more evenly across MPI ranks

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REGION 2

call_path: "Invisible call tree root/funwave_tvd/cal_dispersion"

severity: 4.36%

slowest_ranks: [8,0,6]

reason: The runtimes vary modestly, with the slowest rank about 4% slower than the average, indicating a mild load imbalance.

investigate: Compare per-rank computational work and iteration counts for ranks 8, 0, and 6 to identify why these ranks take longer.

fix: Redistribute a portion of the mesh or workload from rank 8 to other ranks to equalize compute time.

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REGION 3

call_path: "Invisible call tree root/funwave_tvd/fluxes/construction_ho/construct_ho_x"

severity: 3.53%

slowest_ranks: [8]

reason: The runtimes are clustered around 14.8 s with one rank taking about 15.36 s, indicating a modest imbalance of roughly 3.5 %.

investigate: compare the number of mesh cells or computational elements assigned to rank 8 with those of the other ranks to see if it owns more work.

fix: rebalance the mesh partitions so that rank 8 receives fewer cells, equalizing the workload across ranks.

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REGION 4

call_path: "Invisible call tree root/funwave_tvd/fluxes/construction_ho/construct_ho_y"

severity: 6.25%

slowest_ranks: [8]

reason: The runtimes vary between ~24.4 s and ~27.6 s, indicating a moderate imbalance where the slowest rank is about 6 % slower than the average.

investigate: compare the amount of mesh or data elements processed by rank 8 versus the faster ranks to verify that it carries more work.

fix: rebalance the mesh partitions so that rank 8 receives a smaller portion, equalizing the workload across ranks.

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REGION 5

call_path: "Invisible call tree root/funwave_tvd/estimate_huv/get_eta_u_v_hu_hv/tridx/MPI_Wait"

severity: 2.26%

slowest_ranks: [0,1]

reason: The runtimes vary by only about 2 % from the average, indicating a slight but not severe imbalance.

investigate: compare the MPI_Wait durations and pending request counts on ranks 0 and 1 against the faster ranks to see if they are waiting on more communications.

fix: overlap communication with computation or redistribute work so that ranks 0 and 1 have fewer outstanding MPI requests.

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REGION 6

call_path: "Invisible call tree root/funwave_tvd/output/preview/putfile/MPI_Barrier"

severity: 4.4%

slowest_ranks: 1

reason: The barrier runtime shows a modest imbalance, with rank 1 taking the longest time while rank 0 finishes considerably faster, indicating uneven preceding work rather than the barrier itself.

investigate: compare the per-rank work and synchronization points just before the MPI_Barrier, focusing on why rank 1 incurs more time than the others.

fix: redistribute the computational load that precedes the barrier so that rank 1 performs comparable work to the rest, reducing its extra idle time.

Finished analyzing all selected regions.

2 – 6 error and shut CubeGUI instantly after hitting the analyze button. LLM hasn't been able to process system trees larger than ~8,192 system tree ranks.

The largest system tree size it's worked for so far is 8,192.

- 2. Path: 3/NEST/bench_strong/scorep_nest_1p1024xO_sum → profile.cubex**
Time taken:
10 roots; 43 call paths, 16,384 system tree elements
- 3. Path: 3/NEST/bench_strong/scorep_nest_1p2048xO_sum → profile.cubex**
Time taken:
10 roots; 43 call paths, 32,768 system tree elements
- 4. Path: 3/NEST/bench_strong/scorep_nest_1p4096xO_sum → profile.cubex**
Time taken:
10 roots; 43 call paths, 65,536 system tree elements
- 5. Path: 3/NEST/bench_strong/scorep_nest_1p8192xO_sum → profile.cubex**
Time taken:
10 roots; 43 call paths, 131,072 system tree elements
- 6. Path: 3/NEST/bench_strong/scorep_nest_1p16384xO_sum → profile.cubex**
Time taken:
10 roots; 43 call paths, 262,144 system tree elements
- 7. Path: 3/NEST/bench_strong/scorep_nest_1p28672xO_sum → profile.cubex**
Time taken:
roots; call paths, system tree elements

The profile doesn't even open on my laptop. GUI not responding error when I try opening this profile

All (8 - 13) error and shut CubeGUI instantly after hitting the analyze button. LLM hasn't been able to process system trees larger than ~8,192 system tree ranks.

The largest system tree size it's worked for so far is 8,192.

- 8. Path: 3/NEST/bench_weak2/scorep_nest_1p1024xO_sum**
Time taken:
10 roots; 43 call paths, 16,384 system tree elements

9. Path: 3/NEST/bench_weak2/scorep_nest_1p2048xO_sum

Time taken:

10 roots; 43 call paths, 32,768 system tree elements

10. Path: 3/NEST/bench_weak2/scorep_nest_1p4096xO_sum

Time taken:

10 roots; 43 call paths, 65,536 system tree elements

11. Path: 3/NEST/bench_weak2/scorep_nest_1p8192xO_sum → profile.cubex

Time taken:

10 roots; 43 call paths, 131,072 system tree elements

12. Path: 3/NEST/bench_weak2/scorep_nest_1p16384xO_sum → profile.cubex

Time taken:

10 roots; 43 call paths, 262,144 system tree elements

13. Path: 3/NEST/run3/scorep_nest_1p28672xO_sum → profile.cubex

Time taken:

roots; call paths, system tree elements

The profile doesn't even open on my laptop. GUI not responding error when I try opening this profile. Too large and LLM wouldn't be able to analyze as seen from previous data.

Files in the paths below are too large to analyze.

14. Path: 3/NEST/run2 – 3 files

15. Path: 3/NEST/run3 – 2 files

16. Path: 3/NEST/stats